Analysis of Algorithms

Project Proposal: Particle Swarm Optimization

Application: Wi-Fi Access Point Placement

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Implementation: C++

Project Overview

Particle Swarm Optimization (PSO) is a population-based optimization algorithm inspired by the collective behavior of swarms. Each candidate solution, called a particle, moves through the search space according to its current velocity, its own best-known position (*personal best*), and the best solution found by the swarm (*global best*). This project will implement PSO from scratch in C++ and apply it to the optimization of Wi-Fi access point placement in a two-dimensional area.

1. What kinds of problems can this algorithm solve?

PSO is mainly used for continuous optimization problems where solutions can be represented by real-valued variables and evaluated using an objective function. It is useful for nonlinear, multimodal, and derivative-free optimization problems.

2. What are its assumptions?

The initial model assumes a bounded two-dimensional area, known user locations, a fixed number of access points, and a fitness function capable of evaluating each candidate solution. The first version will use a simplified distance-based model without considering walls or signal interference.

3. What would constitute an interesting application?

The selected application is *Wi-Fi access point placement*. Given a set of user locations, PSO will determine suitable access-point coordinates in order to minimize the average distance between each user and its nearest access point.

4. What data representation is appropriate?

Each user will be represented by coordinates (x_i, y_i). A particle will represent the positions of all k access points as

$$(x_1, y_1, x_2, y_2, \ldots, x_k, y_k).$$

Therefore, a problem with k access points will have a $2k$-dimensional search space.

5. What competing approach should be considered?

PSO will primarily be compared with *K-means clustering*, since its cluster centroids provide a natural baseline for a distance-based placement problem. Random Search may also be used as a simple additional baseline. The comparison will focus on solution quality and execution time.

6. What behavior is worth investigating?

The experiments will investigate how swarm size, number of iterations, inertia weight w, cognitive coefficient c_1, and social coefficient c_2 affect convergence and solution quality. The scalability of the algorithm will also be studied by changing the number of users and access points.

Planned Implementation. The algorithm will be implemented entirely in C++. PSO will first be tested using standard benchmark functions such as Sphere and Rastrigin before being applied to the Wi-Fi placement problem. Experimental results will then be used to compare PSO with the selected baseline methods.

Reference: J. Kennedy and R. Eberhart, "Particle Swarm Optimization," *Proceedings of ICNN'95 – International Conference on Neural Networks*, 1995.